Daily AWS Cost Alert Using Cost Explorer API and SNS

Objective

The objective of this assignment is to build an automated AWS cost monitoring solution using AWS Lambda, Amazon EventBridge, Amazon SNS, and the AWS Cost Explorer API. The Lambda function retrieves the current month-to-date AWS spending, compares it against a predefined threshold, and sends an email notification if the threshold is exceeded.

AWS Services Used

- AWS Lambda (Python 3.12)
- Amazon EventBridge
- Amazon SNS
- AWS Cost Explorer API
- AWS IAM
- Amazon CloudWatch Logs

Implementation Steps

Step 1: Created an SNS Topic

- Created a Standard SNS topic named **daily-cost-alert-topic**.
- Created an email subscription.
- Confirmed the subscription using the confirmation email sent by AWS.

Step 2: Created a Lambda execution IAM role. An inline policy was added with the following permissions:

- ce:GetCostAndUsage
- sns:Publish

The SNS permission was scoped only to the required SNS topic following the principle of least privilege.

Step 3: Created a Lambda function with:

- Runtime: Python 3.12
- Architecture: x86_64
- Existing IAM execution role

Step 4: Configured Environment Variables. For testing purposes, the threshold was temporarily set to **0.01 USD** to ensure that the notification email was triggered successfully. After testing, the threshold was restored to the desired production value.

Step 5: Implemented the Lambda Function performing the following tasks:

1. Initializes Cost Explorer and SNS clients using Boto3.
2. Retrieves the current month-to-date Unblended Cost using the Cost Explorer API.
3. Prints the retrieved cost to CloudWatch Logs.
4. Compares the retrieved cost with the configured threshold.
5. Publishes an SNS notification if the threshold is exceeded.
6. Logs the SNS Message ID after successfully publishing the notification.

Step 6: Tested the Lambda Function. The Lambda execution successfully:

- Retrieved the current AWS cost.
- Logged the retrieved cost in CloudWatch Logs.
- Published an SNS notification.
- Returned a successful execution response.

Step 7: Created an EventBridge rule with a daily schedule.Following are the Configuration:

- Schedule Type: Rate Expression
- Expression: rate (1 Day)

The EventBridge rule triggers the Lambda function automatically once every day.

Step 8: CloudWatch Logs confirmed that:

- The current month-to-date AWS cost was retrieved successfully.
- The SNS notification was published successfully.
- The Lambda execution completed without errors.

Screenshots Included

The submission contains screenshots of:

1. create_SNS_topic
2. create_SNS_subscription
3. IAM_config
4. lambda_config
5. log-events-viewer-result
6. create_event_schedule

Conclusion

The solution successfully monitors the AWS account's month-to-date spending using the Cost Explorer API. When the configured threshold is exceeded, the Lambda function automatically publishes an SNS notification. The function is scheduled using Amazon EventBridge and all executions are logged in CloudWatch Logs, providing a complete automated cost monitoring workflow.